

QUESTION 1 (12 MARKS)**Marks**

- (a) Factorise $250 - 2x^3$. 2
- (b) Simplify $\frac{x^2}{4x^2 + 7x + 3} - \frac{x-3}{4x+3}$. 3
- (c) Solve $x(2-x) > -3$. 3
- (d) If $0^\circ \leq \theta \leq 360^\circ$, find all values of θ in the following equation: 2
- $$\sin \theta = -\frac{1}{2}$$
- (e) If a and b are rational and $\frac{1+\sqrt{7}}{3-\sqrt{7}} = a + b\sqrt{7}$, find a and b . 2

QUESTION 2 (12 MARKS) Start this question on a new page.

- (a) Solve simultaneously: 3
- $$2x - y = 1$$
- $$y = \frac{3}{x}$$
- (b) Find the exact value of $\cos \beta$ given that $\sin \beta = \frac{4}{9}$ and $\cos \beta < 0$. 2
- (c) (i) On the same diagram, sketch $y = x$ and $y = |x+2|$. 3
- (ii) For what values of x is $|x+2| < x$?
- (d) Sketch each of the following functions 4
- (i) $y = \sqrt{9-x^2}$ (ii) $f(x) = 4 - |1-x|$

Marks

7

(i) Show that $BP = h$ and $DP = h\sqrt{3}$.

(ii) By the use of the Sine Rule in triangle BDP , show that angle $BDP = 30^\circ$ and hence that triangle BDP is right angled at P .

(iii) Prove that $x = 2h$.

(iv) By the use of the Cosine Rule in triangle PDF , show that $PF = h\sqrt{13}$ and hence show that the angle of elevation of E from P is approximately 15.5° .

(b) Find the point that divides the interval from $(-2,1)$ to $(1,3)$ **externally** in the ratio 3:2.

(c) Prove that $\frac{\sec\theta + \operatorname{cosec}\theta}{\tan\theta + \cot\theta} = \sin\theta + \cos\theta$.

QUESTION 4 (12 MARKS) Start this question on a new page.**Marks**

(a) For the function:

3

$$f(x) = \begin{cases} x+2, & x < -2 \\ 2, & -2 \leq x \leq 2 \\ x-2, & x > 2 \end{cases}$$

(i) Sketch $f(x)$ (ii) Evaluate $2f(-2) - f(2) + 3f(4)$.

(b)

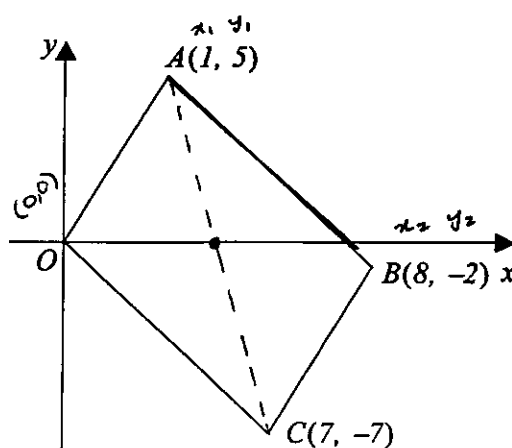
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DIAGRAM NOT DRAWN TO SCALE

In the diagram $O(0, 0)$, $A(1, 5)$, $B(8, -2)$ and $C(7, -7)$ are the vertices of quadrilateral $OABC$.

- (i) Find the midpoint of the interval joining AC .
- (ii) Find the gradient of AB .
- (iii) Show that the equation of AB is $x + y = 6$.
- (iv) Find the exact length of AB .
- (v) Show that AB is parallel to OC .
- (vi) Explain why $OABC$ is a parallelogram.
- (vii) Find the exact perpendicular distance from O to AB .
- (viii) Hence find the area of parallelogram $OABC$.

QUESTION 5**(12 MARKS) Start this question on a new page.****Marks**

(a) For the function $y = \frac{4x}{2x-1}$: 6

- (i) Find the equation of the vertical asymptote.
- (ii) Find the equation of the horizontal asymptote.
- (iii) Find the coordinates of any intercepts for this curve.
- (iv) Sketch the curve, showing all the above information.
- (v) Hence or otherwise, solve $\frac{4x}{2x-1} < 1$.

(b) Sketch the region bounded by the following inequations, the coordinates of the intersections of boundaries are to be shown: 4

$$y > 3x$$

$$y \leq 4 - x^2$$

(c) Find the exact value of $\sec(-210)^\circ$. 2

QUESTION 6**(12 MARKS) Start this question on a new page**

(a) In a rectangular co-ordinate system with scale 1 cm = 1 unit on each axis, we have $A(0, 8)$, $B(6, 0)$, $C(x, 12)$. If $0 < x < 6$ and area $\triangle ABC$ is 20 cm^2 , determine the value of x . 5

(b) Solve $|x-2| + |x+2| = 4$ 3

(c) Find all possible values of θ (to the nearest minute) if $6 \cos^2 \theta + \sin \theta = 5$ and $0 \leq \theta \leq 360^\circ$. 4

QUESTION 7**(12 MARKS)****Start this question on a new page.****Marks**

- (a) A yacht sailing due west, turns at A to avoid a treacherous reef and sails on a course bearing $212^{\circ}20'$ for 2.8 nautical miles to B . It then turns and sails on a course bearing $330^{\circ}35'$ to a point C , due west of A . 4
- (i) Draw a diagram, showing the information above.
- (ii) Find to the nearest tenth of a nautical mile, the distance BC .
- (iii) How much further (to the nearest nautical mile) did the yacht have to sail than if it had maintained its original course?
- (b) If $0^{\circ} \leq \theta \leq 360^{\circ}$, find all values of θ for $\tan 2\theta = \frac{1}{\sqrt{3}}$. 2
- (c) Solve for x : $2 \sin x + \tan x = 0$ for $0 \leq x \leq 180^{\circ}$. 3
- (d) Find all real x such that $\left| \frac{x}{2} \right| > \sqrt{x^2 - 9}$. 3

END OF PAPER

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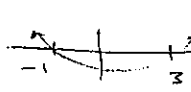
C

C

Question 1

$$\begin{aligned} \text{a) } 250 - 2x^3 &= 2(125 - x^3) \\ &= 2(5-x)(25 + 5x + x^2) \end{aligned}$$

$$\begin{aligned} \text{b) } \frac{x^2}{4x^2+7x+3} - \frac{x-3}{4x+3} &= \frac{x^2}{(4x+3)(x+1)} - \frac{x-3}{4x+3} \\ &= \frac{x^2 - (x-3)(x+1)}{(4x+3)(x+1)} \\ &= \frac{x^2 - (x^2 - 2x - 3)}{(4x+3)(x+1)} \\ &= \frac{2x+3}{(4x+3)(x+1)} \end{aligned}$$

$$\begin{aligned} \text{c) } x(2-x) &> -3 \\ 2x - x^2 &> -3 \\ 0 &> x^2 - 2x - 3 \\ 0 &> (x-3)(x+1) \end{aligned}$$


$$\therefore -1 < x < 3$$

$$\begin{aligned} \text{d) } \sin \theta &= -\frac{1}{2} \end{aligned}$$

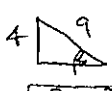

related $\theta = 30^\circ$ 3rd + 4th Q.

$$\therefore \theta = 210^\circ, 330^\circ$$

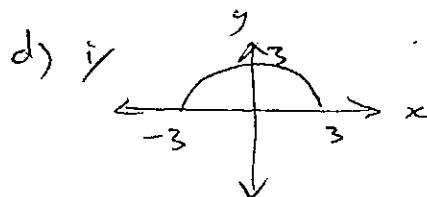
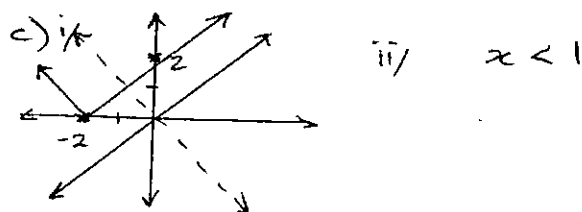
$$\begin{aligned} \text{e) } a + b\sqrt{7} &= \frac{1+\sqrt{7}}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}} \\ &= \frac{3+4\sqrt{7}+7}{-7} \\ &= 5+2\sqrt{7} \end{aligned}$$

Question 2

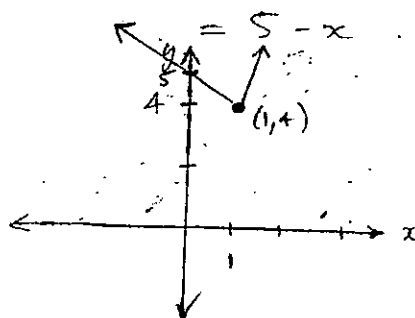
$$\begin{aligned} \text{a) } 2x - \frac{3}{x} &= 1 \\ 2x^2 - 3 &= x \\ 2x^2 - x - 3 &= 0 \\ (2x-3)(x+1) &= 0 \\ x &= \frac{3}{2} \text{ or } -1 \\ \therefore y &= 2 \text{ or } -3 \\ \therefore \left(\frac{3}{2}, 2\right) \text{ or } (-1, -3) \end{aligned}$$

$$\begin{aligned} \text{b) } \sin \beta &= \frac{4}{9} \quad \cos \beta < 0 \\ \cos \beta &= -\frac{\sqrt{65}}{9} \end{aligned}$$


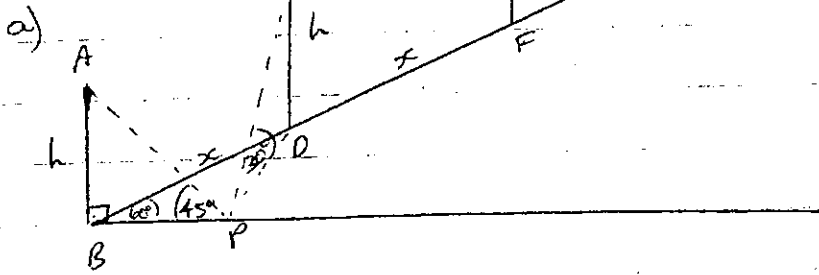
$$\sqrt{9^2 - 4^2} = \sqrt{65}$$



$$\begin{aligned} \text{ii/ } f(x) &= 4 - 1 + x \text{ for } x \geq 1 \\ &= 3 + x \\ f(x) &= 4 + 1 - x \text{ for } x < 1 \end{aligned}$$



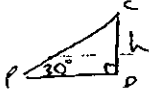
Question 3



i) $\angle APB = \angle PAB = 45^\circ \therefore \text{isose}$

$\therefore BP = h$

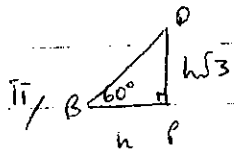
In $\triangle CDP$



$\frac{h}{DP} = \tan 30^\circ$

$\frac{h}{DP} = \frac{1}{\sqrt{3}}$

$\therefore DP = h\sqrt{3}$



$\frac{L}{\sin \angle BDP} = \frac{h\sqrt{3}}{\sin 60^\circ}$

$\frac{\sqrt{3}}{2} = \sqrt{3} \sin \angle BDP$

$\sin \angle BDP = \frac{1}{2}$

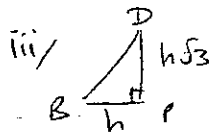
$\angle BDP = 30^\circ \text{ or } 150^\circ (\text{L sum})$

$\therefore$ By angle sum

$\angle P = 180 - (60 + 30)$

$= 90^\circ$

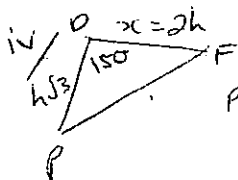
$\therefore$ Right angled @ P.



$BD = \sqrt{h^2 + 3h^2}$

$= \sqrt{4h^2}$

$= 2h$

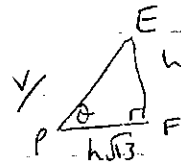


$PF^2 = (2h)^2 + (h\sqrt{3})^2 - 2 \cdot 2h \cdot h\sqrt{3} \cos 150^\circ$

$= 4h^2 + 3h^2 - 4\sqrt{3}h^2 \cdot -\frac{\sqrt{3}}{2}$

$= 13h^2$

$\therefore PF = h\sqrt{13}$



$\tan \theta = \frac{h}{h\sqrt{3}}$

$\theta = \tan^{-1}(\frac{1}{\sqrt{3}})$

$\theta = 15.5^\circ$

b) $(-2, 1), (1, 3) \quad -3 = 2$

$x = \frac{2x - 2 + -3 \times 1}{-3 + 2}$

$= 7$

$y = \frac{2 \times 1 - 3 \times 3}{-3 + 2}$

$= 7$

$(7, 7)$

c) $\frac{\sec \theta + \csc \theta}{\tan \theta + \cot \theta} = \sin \theta + \cos \theta$

LHS = $\frac{\frac{1}{\cos \theta} + \frac{1}{\sin \theta}}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}} \times \frac{\sin \theta}{\sin \theta}$

$= \frac{\sin \theta + \cos \theta}{\sin^2 \theta + \cos^2 \theta}$

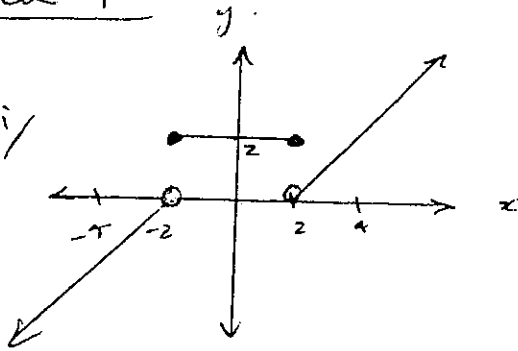
$= \sin \theta + \cos \theta$

$= \text{RHS}$

$\therefore$ True.

Question 4

a) i/



$$\text{ii/ } f(-2) = 2$$

$$f(2) = 2$$

$$f(-4) = -4 + 2 \\ = -2$$

$$\therefore 2 \times 2 - 2 + 3 \times -2 \\ = 4 - 2 - 6 \\ = -4$$

b) i/ $M_{AC} = \left(\frac{1+7}{2}, \frac{5-7}{2} \right)$
 $= (4, -1)$

ii/ $M_{AB} = \frac{-2-5}{8-1}$
 $= \frac{-7}{7}$
 $= -1$

iii/ eqn AB $\Rightarrow y - 5 = -1(x - 1)$
 $y - 5 = -x + 1$
 $x + y - 6 = 0$
 $\therefore x + y = 6$

iv/ $d_{AB} = \sqrt{(8-1)^2 + (-2-5)^2}$
 $= \sqrt{49 + 49}$
 $= 7\sqrt{2}$

v/ $M_{OC} = \frac{-7}{7}$
 $= -1$

$\therefore$ Since $M_{AB} = M_{OC}$
 $AB \parallel OC$

vi/ $d_{OC} = \sqrt{7^2 + 7^2}$
 $= 7\sqrt{2}$

$\therefore OABC$ is a //gm
 $\therefore$ it has a pair of equal & parallel side

vii/ $(0,0)$ $x + y - 6 = 0$
 $d = \frac{|1 \times 0 + 1 \times 0 - 6|}{\sqrt{1+1}}$
 $= \frac{6}{\sqrt{2}}$

$d = 3\sqrt{2}$

viii/ $A = AB \times 3\sqrt{2}$
 $= 7\sqrt{2} \times 3\sqrt{2}$
 $= 21 \times 2$
 $A = 42 \text{ u}^2$

Question 5

a) $y = \frac{4x}{2x-1}$

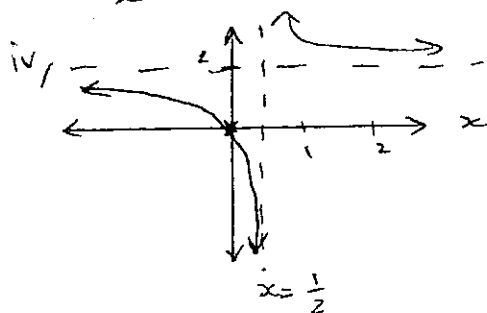
i, $x = \frac{1}{2}$

ii, $\lim_{x \rightarrow \infty} \frac{4x}{2x-1}$

$= \frac{4}{2}$

$y = 2$

iii, $y \text{ int} \rightarrow 0$
 $x \text{ int} \rightarrow 0$



v, when $y = 1$

$1 = \frac{4x}{2x-1}$

$2x-1 = 4x$

$-1 = 2x$

$x = -\frac{1}{2}$

$\therefore x < -\frac{1}{2} \text{ or } x > \frac{1}{2}$

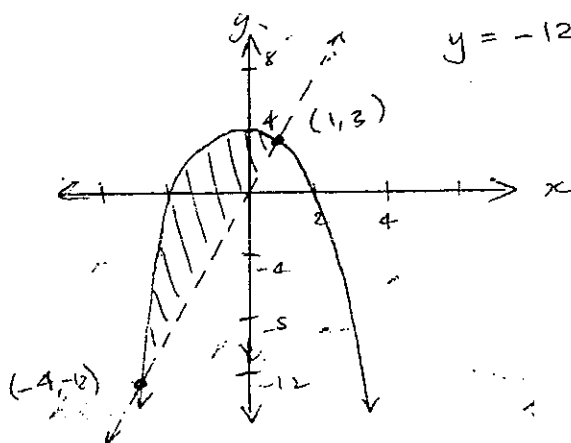
b) Intersection: $3x = 4 - x^2$

$x^2 + 3x - 4 = 0$

$(x+4)(x-1) = 0$

$x = -4, 1$ $(-4, 12)$

$y = -12, 3$ $(1, 3)$

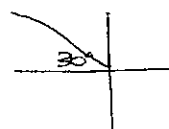


c) $\sec(-210)^\circ$

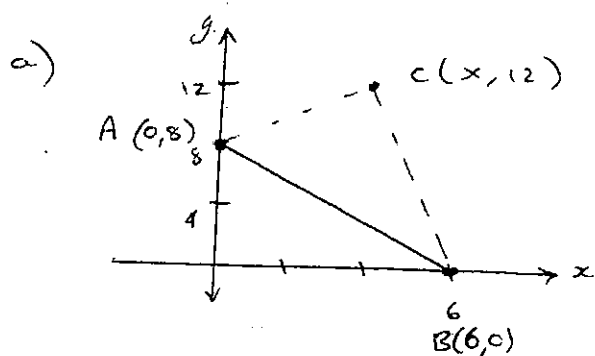
$= \frac{1}{\cos(-210)^\circ}$

$= \frac{1}{-\frac{\sqrt{3}}{2}}$

$= -\frac{2}{\sqrt{3}}$



Question 6



$$\text{Area } \triangle ABC = 20 \text{ cm}^2 \quad 0 < x < 6$$

$$AB = \sqrt{8^2 + 6^2}$$

$$= 10 \text{ cm}$$

Perp Ht $\Rightarrow C(x, 12)$

$$AB: y = -\frac{4}{3}x + 8$$

$$4x + 3y - 24 = 0$$

$$= \frac{|4x + 36 - 24|}{\sqrt{4^2 + 3^2}}$$

$$= \frac{|4x + 12|}{5}$$

$$\text{Area} = \frac{1}{2} \times 10 \times \frac{|4x + 12|}{5}$$

$$20 = |4x + 12|$$

$$\therefore 4x + 12 = 20 \quad \text{or} \quad -4x - 12 = 20$$

$$4x = 8$$

$$x = 2$$

$$-4x = 32$$

$$x = -8 \times$$

No negative distances.

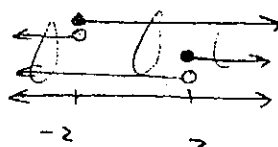
b) $|x-2| + |x+2| = 4$

$$|x-2| = x-2 \quad \text{for } x \geq 2$$

$$-x+2 \quad \text{for } x < 2$$

$$|x+2| = x+2 \quad \text{for } x \geq -2$$

$$-x-2 \quad \text{for } x < -2$$



Case ① $-x+2 -x-2 = 4 \quad \text{for } x < -2$

$$-2x = 4$$

$$x = -2$$

② $-x+2 + x+2 = 4 \quad \text{for } -2 \leq x < 2$

$$4 = 4$$

$$\therefore \text{all } x \quad -2 \leq x < 2$$

③ $x-2 + x+2 = 4$

$$2x = 4$$

$$x = 2$$

$$\therefore \text{Solution: } -2 \leq x \leq 2$$

c) $6\cos^2\theta + \sin\theta = 5 \quad 0 \leq \theta \leq 360^\circ$

$$6(1-\sin^2\theta) + \sin\theta = 5$$

$$6 - 6\sin^2\theta + \sin\theta = 5$$

$$6\sin^2\theta - \sin\theta - 1 = 0$$

$$(3\sin\theta + 1)(2\sin\theta - 1) = 0$$

$$\sin\theta = -\frac{1}{3} \quad \text{or} \quad \sin\theta = \frac{1}{2}$$

$$\text{related } \theta = 19^\circ 28'$$

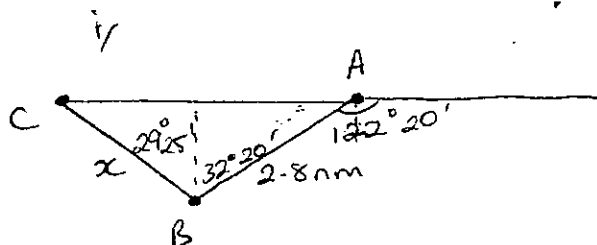
$$\theta = 30^\circ, 150^\circ$$

$$\theta = 199^\circ 28', 340^\circ 32'$$

$$\therefore \theta = 30^\circ, 150^\circ, 199^\circ 28', 340^\circ 32'$$

Question 7

a)



$$\begin{aligned} \text{ii/ } \angle ABC &= 29^\circ 25' + 32^\circ 20' \\ &= 61^\circ 45' \\ \angle BAC &= 57^\circ 40' \\ \therefore \angle ACB &= 60^\circ 35' \end{aligned}$$

$$\text{iii/ } AC = \frac{2.8 \sin 61^\circ 45'}{\sin 60^\circ 35'} = 2.83$$

$$AB + BC = 5.51 \text{ nm}$$

$$\therefore 5.51 - 2.83 = 2.686 \text{ nm}$$

$\therefore$ Yacht travels 2.7 nm further.

$$\begin{aligned} \therefore \frac{x}{\sin 57^\circ 40'} &= \frac{2.8}{\sin 60^\circ 35'} \\ x &= \frac{2.8 \sin 57^\circ 40'}{\sin 60^\circ 35'} \end{aligned}$$

$$\therefore BC = 2.7 \text{ nm}$$

$$\text{b) } \tan 2\theta = \frac{1}{\sqrt{3}} \quad 0 \leq \theta \leq 360^\circ$$

$$\tan u = \frac{1}{\sqrt{3}} \quad 0 \leq \theta \leq 720^\circ$$

$$\text{related } u = 30^\circ$$

$$* \quad u = 30^\circ, 210^\circ, 390^\circ, 570^\circ$$

$$\therefore \theta = 15^\circ, 105^\circ, 195^\circ, 275^\circ$$

$$\text{c) } 2 \sin x + \tan x = 0 \quad 0 \leq x \leq 180^\circ$$

$$2 \sin x + \frac{\sin x}{\cos x} = 0$$

$$\sin x \left(2 + \frac{1}{\cos x} \right) = 0$$

$$\sin x (2 + \sec x) = 0$$

$$\therefore \sin x = 0 \quad \text{or} \quad \sec x = -2$$

$$x = 0, 180^\circ \quad \cos x = -\frac{1}{2}$$

$$x = 120^\circ$$

$$\therefore x = 0, 120^\circ, 180^\circ$$

$$\text{d) } \left| \frac{x}{2} \right| > \sqrt{x^2 - 9}$$

square both sides.

$$\frac{x^2}{4} > x^2 - 9$$

$$x^2 > 4x^2 - 36$$

$$36 > 3x^2$$

$$x^2 > 12$$

$$x^2 - 12 > 0$$

$$(x - 2\sqrt{3})(x + 2\sqrt{3}) > 0$$

$$\therefore x < 2\sqrt{3} \quad \text{or} \quad x > 2\sqrt{3}$$

check

$$\sqrt{x^2 - 9} \quad x^2 - 9 \geq 0$$

$$(x - 3)(x + 3) \geq 0$$

$$x \geq 3 \quad \text{or} \quad x \leq -3$$

$\therefore$ solution is fine